

Writing a Good Mathematical Paper

A compact guide for authors and AI assistants

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A reader who knows the field should be able to state the result, reconstruct the decisive inference and use it correctly. Begin where that reader would get stuck; use the relevant instruction, not every instruction.

1. Decide what the paper contributes. Choose a coherent principal result and an intended reader. Explain what the result enables or clarifies, compare its assumptions and conclusion with the nearest prior work, and state what remains open. For a conditional result, say which familiar cases its hypothesis admits or excludes, where known. Distinguish the contribution from the mechanism that proves it; the abstract needs both. Neither theorem count nor generality establishes importance.

2. Read the kind of mathematics you are writing. Read nearby papers for terminology, hypotheses and attribution; read complete local arguments for explanation. Match the task: a parameter choice, limit or estimate. Check the version and passage. Identify what justifies the source's next step and the corresponding fact in your proof; explain it in your own words.

3. Check the statement before polishing its prose. Check domains, quantifiers, signs, endpoints, constants, parameter dependencies and both directions of any claimed equivalence. State assumptions where the claim is made. Compare titles, abstracts and conclusions with the full statement. Separate witnesses need not be a common witness: over the natural numbers, $\forall n \exists x (x > n)$ does not imply $\exists x \forall n (x > n)$. An exposition pass that exposes a proof gap must refer it for mathematical review; elegant wording cannot close it.

4. Organise by logical dependence. Give the question, result and proof idea before technical detail accumulates. Show what each major step establishes and why the next needs it. Introduce prerequisites when their purpose is visible, not all at once. Prepare unfamiliar operations with examples. Let headings name the object, result or operation; make assumptions and proofs findable from the theorem. Logical order need not reproduce discovery order.

5. Explain the hard transition. Work backwards from the next conclusion: what quantity must be controlled, and how accurately? State that requirement before deriving the estimate or choosing a parameter. For an interval certificate, compare the error with both endpoint gaps. For a small-integer contradiction, justify integrality, nonvanishing and $|N| < 1$ separately. When a technique recurs, identify its new job. Explain the decisive inference, not merely the calculation being performed.

6. Separate construction from completion. Account for each choice and preserved condition. A classification needs both construction and exhaustiveness. An enclosure alone does not show that every target is attained. For example, $r_n = x - S_n$, $0 \leq r_n \leq \varepsilon_n$ and $\varepsilon_n \rightarrow 0$ give $S_n \rightarrow x$; repeated choice alone does not. State which parameters are fixed and justify the limit of the whole summand, including any growing product. To interchange a limit and a sum, identify an applicable theorem and check its hypotheses.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

7. Make examples and figures do work. Choose a small instance that exposes a definition, invariant, obstruction or necessary hypothesis; a simpler case may hide it. Show what changes and what is preserved. Include an endpoint when the convention matters, and say what the example leaves unproved. A finite check supports an infinite claim only through a proved reduction. Check that a witness belongs to the claimed object. Label a diagram's objects, maps and conventions; keep its explanation nearby.

8. Use the field's words and purposeful notation. Keep established terms when their definitions fit, such as "Stieltjes moment sequence". Replace private labels by the actual condition; $u_n/2^n \rightarrow 0$ is weaker than subexponential growth. Define symbols by their role before use and keep that role stable. Remove abbreviations whose translation costs more than their reuse saves. Translate borrowed indices and normalisations. Distinguish a sum, its partial sums and its remainders, and a value from its enclosure.

9. Make sentences express the argument. Replace "We proceed with the estimate" by a reason: "Orthogonality makes the first sum zero; it remains to bound the second." Let "since" supply a reason and "hence" mark a deduction. Give each "this" an unmistakable referent. After "similarly", check that the same hypotheses apply. Punctuate displays as parts of sentences. Keep parallel claims parallel and conditions beside their consequences. Remove filler, not mathematical qualifications.

10. Credit the exact input at its use. Distinguish applied theorems, adapted constructions, historical attribution, terminology and expository models. Check hypotheses, version and passage; identify the input used here. State what an adaptation retains and changes. Credit supplied ideas and corrections locally. A writing analogy supplies no proof. Report only reading and checks performed.

11. State what has actually been established. Separate logical scope from evidence: conditional implications can have ordinary or formal proofs; computations and conjectures need separate labels. Keep unproved inputs beside dependent results. A checked lemma need not cover the whole proof; missing formalisation does not invalidate an ordinary proof. State computations' finite scope locally; collect reproduction details in one verification section. For one-sided tests, distinguish success from inconclusive failure. Pointwise eventual detection need not give a uniform stopping bound or the required family of witnesses.

12. Preserve useful detail and informative failures. Keep the short argument readable and omitted detail recoverable. Check exact passage destinations and reconcile statements, assumptions and evidence in both directions. Preserve a useful failed route's implication, assumptions, witness and missing ingredient. Distinguish flawed steps, failed methods and counterexamples. Cite omitted proofs by section or theorem; put build commands and source identifiers in the verification record. Do not invent discovery history.

13. Review changes as claims, then as prose. Compare meanings before fluency, including words beside unchanged formulas. Correct overclaims and stale underclaims. Removing shorthand must preserve its hypotheses; terminology must fit its definition. Accept, repair, reject or defer a proposal for a stated reason. Keep useful rejected revisions recoverable. Review scope and limits separately before turning a local improvement into a general lesson.

14. Read the delivered paper from a cold start. Render the final sources; inspect formulas, captions, page breaks, bibliography and reference destinations. Read without supplying steps from memory. Can the reader reconstruct the hard inference, apply the result and identify an excluded case, where known? Record where reconstruction stops and what was understood there, not just that the passage feels unclear. A successful build is not a comprehension test. Revise the difficulty; preserve what works.

Further reading. Halmos, *How to write mathematics*, §§3–5 (1970); Knuth et al., *Mathematical Writing*, §§1–3 (1988); Gowers, *Examples first!* (2007); Tao, *On writing*. Worked cases, sources and revision history: the *long companion*.